# Résumé: Dr. Matthias Mittner (born Ihrke)

## Contact Information

Matthias Mittner (born Ihrke)

*University of Tromsø*
Huginbakken 32
9037 Tromsø
Norway

Phone: +47 776 46371 (office)
E-mail: `matthias.mittner@uit.no`
Web: `http://ihrke.github.io/`


## Personal Information

Date of birth: September 1, 1982
Place of birth: Eberswalde, Germany
Nationality: german
Marital Status: Married to Lilli Mittner
Children: Kuno, Josefine & Klara Mittner

## Education

*Dr. phil. (summa cum laude)* — 21.12.2011
University of Frankfurt, Germany
Thesis: "Binding and Retrieval of Stimulus-Response Episodes in the Negative Priming Paradigm"

*B. Sc. in Computer Science* — 2005/07 – 2008/04
Institute of Computer Science, University of Göttingen, Germany
Thesis: "Single Trial Estimation and Timewarped Averaging of Event-Related Potentials"

*Diploma in Psychology* — 2005/07 – 2007/11
Department of Psychology, University of Göttingen, Germany
Thesis: "Negative Priming and Response-Relation: Behavioural and Electroencephalographic Correlates"

*Student at the University of Oslo* — 2004/08 – 2005/06
Oslo, Norway
Erasmus-Scholarship; studies of computer science and psychology

*B. Sc. in Psychology (equivalent)* — 2002/09 – 2004/07
Department of Psychology, University of Göttingen, Germany
Thesis: "An Adaptive Approach to Memory-Span Tasks"

*High-School Diploma* — 2002/06
Alexander von Humboldt Gymnasium, Eberswalde, Germany
main subjects physics and mathematics

## Employments

*Associate Professor* — 2014/04 – present
Institute of Psychology, University of Tromsø, Tromsø, Norway
Cognitive Neuroscience

*Post-Doctoral Researcher* — 2013/04 – 03/2014
Cognitive Science Center Amsterdam, Universiteit van Amsterdam, The Netherlands

Cognitive Neuroscience (Birte Forstmann)

*Post-Doctoral Researcher* 2012/09 – 2013/03
Max-Planck Institute for Experimental Medicine, Göttingen, Germany
Theoretical Neuroscience (Robert Gütig)

*PhD-Student* 2008/10 – 2012/01
Max-Planck Institute for Dynamics and Self-Organization, Göttingen, Germany
Göttingen Graduate School for Neurosciences and Molecular Biosciences (GGNB)
Doctoral Program: Theoretical and Computational Neuroscience (PTCN)

*Guest Researcher* 2008/03 – 2008/09
Max-Planck Institute for Dynamics and Self-Organization, Göttingen, Germany
cognitive modelling and data-mining (EEG)

*Guest Researcher* 2007/11 – 2008/02
Max-Planck Institute for Informatics, Saarbrücken, Germany
section for computer graphics; optical modelling and psychophysics

*Research Assistant* 2005/09 – 2007/10
Bernstein Center for Computational Neuroscience, Göttingen, Germany
cognitive modelling; psychophysical experiments

*Student Assistant* 2003/09 – 2004/08
Departement for Developmental Psychology, University of Göttingen, Germany
computer administration and development of experimental software

## Scientific Activities

*Leadership*
Leader of research group "Cognitive Neuroscience" at the institute for psychology, university of Tromsø (2018–)
`http://www.uit.no/research/cognitive-neuroscience`

*Professional Memberships*
Member of the Society for Neuroscience
Program Committee Member "International Conference on Intelligent Data Engineering and Automated Learning" 2010 and 2011
Member of the Research Priority Program Brain and Cognition at the Cognitive Science Center Amsterdam (CSCA)
Review Editor for "Frontiers in Psychology"

*Committee*
IDEAL 2010/2011 (program-committee)
Evaluation committee for several PhD and post-doc positions (IPS/UiT)
Member of "Master committee" (responsible for master studies at IPS/UiT; 2015–2016)
Vararepresentant programstyret psykologi (IPS/UiT; 2017–)
Committee leader for PhD defence of Sara Vambheim (2018)
Committee development of evaluation strategy at IPS (UiT, 2018–)
Committee for selection of student price for best thesis (clinical program, 2018–)

*Advisor*
Netherlands Organisation for Scientific Research (NWO) - Vici Social Sciences and Humanities
Hertie Foundation Project: Improvement of school education by neuroscientific research

*Organizing*

Organizer for Methods-course "Introduction to Python programming" (2012)
Organizer for Methods-course "Matlab and Python programming" (2011)

*Conferences*
Invited presentations:
– The Role of Neural Oscillations in Human Cognition (Summer School, Uslar, Germany), 2017
– 12th International Conference on Cognitive Neuroscience, Brisbane, Australia (ICON 2014)
Various talks and posters at national and international conferences (CNS 06/07, BCCN symposia, TeaP, ICP2008, IDEAL, IJMC, NWG, SfN, MathPsych, ICON2014, ICON2017)

*Advanced Courses*
Program for young research leaders 2014–2015; joint program UiO, UiB, NTNU, UiT – leadership, networking, financing, communication, strategy

DGPs Course on Recording and Analysis of Eye-Movements (Niels Galley), Cologne 2008

Course on Computational Neuroscience at the Max-Planck-Institute for Dynamics and Self-Organization 2005

*Reviewing*
Reviewer for Spring Conference on Computer Graphics 2008 (SCCG08)
Ad-hoc Reviewer for Cognitive Neuropsycholoy, NeuroImage, Journal of Cognitive Neuroscience, Social Cognitive & Affective Neuroscience, Frontiers in Psychology
IDEAL 2010/2011 (PC-member)

## Publications

*Accepted/pre-print/submitted*

(35) Turi, Z., Bjørkedal, E., Gunkel, L., Antal, A., Paulus, W., & *Mittner, M.* (2018, May 23). Evidence for Cognitive Placebo and Nocebo Effects in Healthy Individuals. `http://doi.org/10.17605/OSF.IO/87KEA`

(34) Hawkins, G., *Mittner, M.*, Forstmann, B. & Heathcote, A. (submitted). Modeling distracted performance.

(33) Boayue, N., Csifcsak, G., Aslaksen, P., Turi, Z., Antal, A., Groot, J., Hawkins, G., Forstmann, B., Opitz, A., Thielscher, A. and *Mittner, M.* (in-principle acceptance). Increasing propensity of mind wandering by transcranial direct current stimulation: A registered report. *European Journal of Neuroscience.*

*Journal Articles*

(32) Hetland A., Vittersø J., Wie S.O.B., Kjelstrup E., *Mittner M.* & Dahl T.I. (2018) Skiing and Thinking About It: Moment-to-Moment and Retrospective Analysis of Emotions in an Extreme Sport. *Frontiers in Psychology.* 9:971. `doi:10.3389/fpsyg.2018.00971`

(31) Boayue N. M., Csifcsák G., Puonti O., Thielscher, A., & *Mittner, M.* (2018) Head models of healthy and depressed adults for simulating the effects of non-invasive brain stimulation [version 1; referees: 1 approved]. *F1000Research.*, 7:704 `doi:10.12688/f1000research.15125.1`, preprint: `psyarxiv.com/e5agy/`.

(30) Turi, Z., Schäfer, S. A., Antal, A., Paulus, W. & *Mittner, M.* (2018). Data from Placebo Enhances Reward Learning in Healthy Individuals. *Journal of Open Psychology Data.* 6 (1), p. 2. `doi:10.5334/jopd.34`

(29) Csifcsák, G., Boayue, N. M., Puonti, O., Thielscher, A., & *Mittner, M.* (2018). Effects of transcranial direct current stimulation for treating depression: A modeling study. *Journal of Affective Disorders.* 234, 164 – 173. `doi:10.1016/j.jad.2018.02.077`, preprint: `psyarxiv.com/erwvu`.

(28) Csifcsák, G. & *Mittner, M.* (2017). Linking brain networks and behavioral variability to different types of mind-wandering. *Proceedings of the National Academy of Sciences.* 114, 30.

(27) Turi*, Z., *Mittner*, M.*, Paulus, W. and Antal, A. (2017). Placebo Intervention Enhances Reward Learning in Healthy Individuals. *Scientific Reports*. 7, 41028. doi:10.1038/srep41028
*shared first authorship

(26) Hawkins, G., *Mittner, M.*, Forstmann, B. & Heathcote, A. (2016). On the efficiency of neurally-informed cognitive models to identify latent cognitive states. *Journal of Mathematical Psychology*.

(25) *Mittner, M.*, Hawkins, G., Boekel, W. & Forstmann, B. (2016). A neural model of mind wandering. *Trends in Cognitive Sciences*. doi:10.1016/j.tics.2016.06.004

(24) Rodriguez-Aranda, C., *Mittner, M.* & Vasylenko, O. (2016). Association between Executive Functions, Working Memory, and Manual Dexterity in Young and Healthy Older Adults: An exploratory study. *Perceptual & Motor Skills*, 122(1), 165–192.

(23) Hawkins, G.E., *Mittner, M.*, Boekel, W., Heathcote, A., and Forstmann, B.U. (2015). Towards a model-based cognitive neuroscience of mind wandering. *Neuroscience*. 310, 290-305.

(22) Høifødt R. S., *Mittner M.*, Lillevoll K. R., Katla S. K., Kolstrup N., Eisemann M., Friborg O., Waterloo K. (2015). Predictors of response to Web-based cognitive behavioral therapy with face-to-face therapist support for depression: A Bayesian analysis. *Journal of Medical Internet Research*, 17(9):e197.

(21) *Mittner, M.*, Boekel, W., Tucker, A. M., Turner, B.M., Heathcote, A. and Forstmann, B.U. (2014). When the brain takes a break: A model-based analysis of mind wandering. *Journal of Neuroscience*, 34(49):16286-95.

(20) Turi*, Z., *Mittner*, M.*, Opitz, A., Popkes, M., Paulus, W., and Antal, A. (2014). Transcranial direct current stimulation over the left prefrontal cortex increases randomness of choice in instrumental learning. *Cortex*, 63:145–154.
*shared first authorship

(19) *Mittner M.*, Behrendt J., Menge U., Titz C. M., Hasselhorn M. (2014). Response-Retrieval in Identity Negative Priming is Modulated by Temporal Discriminability. *Frontiers in Psychology*. 5:621.

(18) *Mittner, M.* (2013). Functional Integration of Large-Scale Brain Networks. *The Journal of Neuroscience*, 33(48):18710-18711

(17) Schrobsdorff H., *Ihrke M.*, Behrendt J., Hasselhorn M., Herrmann J.M. (2012). Inhibition in the Dynamics of Selective Attention: An Integrative Model for Negative Priming. *Frontiers in Psychology* doi:10.3389/fpsyg.2012.00491

(16) *Ihrke M.*, Behrendt J., Schrobsdorff H., Visser I., Hasselhorn M. (2012). Negative Priming Persists in the Absence of Response-Retrieval. *Experimental Psychology*, 60(1):12-21.

(15) Schrobsdorff H., *Ihrke M.*, Behrendt J., Herrmann J. M., Hasselhorn M. (2012). Identity Negative Priming: A Phenomenon of Perception, Recognition or Selection? *PLoS ONE 7* (3), doi:10.1371/journal.pone.0032946.

(14) *Ihrke, M.* & Behrendt, J. (2011). Automatic generation of randomized trial sequences for priming experiments. *Frontiers in Psychology, 2* (255), doi:10.3389/fpsyg.2011.00225.

(13) *Ihrke, M.* & Brennen, T. (2011). Sharing one biographical detail elicits priming between famous names: Empirical and computational approaches. *Frontiers in Psychology*, 2 (75), doi:10.3389/fpsyg.2011.00075.

(12) *Ihrke, M.*, Schrobsdorff, H. & Herrmann, J.M. (2011). Recurrence-Based Estimation of Time-Distortion Functions for ERP Waveform Reconstruction. *International Journal of Neural Systems*, 21 (1), pp. 65-78.

(11) *Ihrke, M.*, Behrendt, J., Schrobsdorff, H., Herrmann, J.M. & Hasselhorn, M. (2011). Response Retrieval and Negative Priming: Encoding and Retrieval Specific Effects. *Experimental Psychology*, 58 (2), pp. 154–161.

(10) Behrendt, J., Gibbons, H., Schrobsdorff, H., *Ihrke, M.*, Herrmann, J.M. & Hasselhorn, M. (2010). Event-Related Brain Potential Correlates of Idendity Negative Priming From Overlapping Pictures. *Psychophysiology*, 47 (5), pp. 921–930.

(9) Ritschel, T., *Ihrke, M.*, Frisvad, J.R., Coppens, J., Myszkowski, K. & Seidel, H.P. (2009). Temporal Glare: Real-Time Dynamic Simulation of the Scattering in the Human Eye. *Computer Graphics Forum.* 28 (2). pp. 183-192.

(8) Ritschel, T., Smith, K., *Ihrke, M.*, Grosch, T., Myszkowski, K., & Seidel, H. P. (2008). 3D Unsharp Masking for Scene Coherent Enhancement. *ACM Transactions on Graphics (Proceedings SIGGRAPH 2008).* 27 (3).

(7) Schrobsdorff, H., *Ihrke, M.*, Kabisch, B., Behrendt, J., Hasselhorn, M., Herrmann, J. M. (2007). A Computational Approach to Negative Priming. *Connection Science*, *19* (3), 203-221.

*Book Chapters*

(6) *Ihrke, M.*, Schrobsdorff, H. & Herrmann, J. M. (2009). Denoising and Averaging Techniques for Electrophysiological Data. In Wennberg, R. & Perez-Velazquez, J. L. (Eds.), *Coordinated Activity in the Brain: measurements and relevance to brain function and behaviour.* Springer.

*Peer-Reviewed Conference Proceedings*

(5) Schrobsdorff H., *Ihrke M.*, Herrmann J.M. (2012). Modeling Structure and Dynamics of Selective Attention. *Advances in Intelligent Systems and Computing* (Biologically Inspired Cognitive Architectures 2012), 196, pp. 287–295.

(4) *Ihrke, M.*, Schrobsdorff, H. & Herrmann, J. M. (2009). Recurrence-Based Synchronization of Single Trials for EEG-Data Analysis. *Lecture Notes in Computer Science* 5788, Intelligent Data Engineering and Automated Learning - IDEAL 2009. pp. 118-125.

(3) *Ihrke M.*, Ritschel, T. , Smith, K., Grosch, T., Myszkowski, K., & Seidel, H. P. (2009). A Perceptual Evaluation of 3D Unsharp Masking. *Proceedings Human Vision and Electronic Imaging XIII.*

(2) *Ihrke, M.*, Schrobsdorff, H. & Herrmann, J. M. (2008). Compensation for Speed-of-Processing Effects in EEG-Data Analysis. *Lecture Notes in Computer Science (Proceedings IDEAL 2008).* 5326, pp. 354-361.

(1) Yoshida, A., *Ihrke, M.*, Mantiuk, R. & Seidel, H. P. (2008). Brightness of the Glare-Illusion. *Proceedings of ACM Symposium on Applied Perception in Graphics and Visualization.* pp. 83-90.

## Teaching

*Lectures and supervision* (University of Tromsø) from 2014

Undergraduate Level
- Methods and Statistics III (PSY-1513)
- Methods and Statistics I (PSY-1511)
- Cognitive Neuroscience (PSY-2547, PSY-2553)
- Affective Neuroscience (PSY-2549)
- Biological Psychology (PSY-1004)
- Psychometrics (PSY-1504)
- Practical Course: Gerontopsychology (PSY-2543)
- Research Practical (PSY-1509/1510)
- Research skills for Master students (PSY-3900)

PhD Courses
- Quantitative Research Methods (HEL-8024)
- Applied Linear Regression (HEL-8030)
- Research Design and Statistics (PSY-8002)
- Quantitative Method in Clinical Health Research (HEL-8017)

– Informal reading club for PhD students and faculty: Bayesian statistics

*Introduction to Neuroscientific Methods and Brain Anatomy* summer 2013

Master-level course (University of Amsterdam)
lecture and practical on electroencephalography

*Linear Regression* (PSY-8012) from spring 2015
PhD course (10 credit points)

*Introduction to Python Programming (A123)* winter term, 2011/2012
GGNB methods course (introductory level), University of Göttingen

*Gestaltung von Lehr-Lernprozessen* summer term, 2010
Introductory seminar to the psychology of learning and teaching for students pursuing teaching certification; University of Göttingen

*Matlab and Python Programming (A109)* winter term, 2010/2011
GGNB methods course (introductory level); University of Göttingen

## Supervision

*Post-Doc supervision*

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| Dr. Csifcsák Gábor (2016–) | "Optimizing transcranial direct current stimulation for treatment of chronic pain and depression" (HelseNord, PFP1237_15) |

*PhD students*

| | |
|---|---|
| Nya Mehnwolo Boayue (2016–) | "Neural and computational correlates of mind wandering" (main supervisor) |
| Isabel Viola Kreis | "Too precise or too imprecise: which parameter is gone awry in autism and psychosis", University of Tromsø (co-supervision, main supervisor: Gerit Pfuhl) |
| Zsolt Turi (2011–2015) | "Methodological and cognitive aspects of transcranial electrical stimulation", University of Göttingen (co-supervision with Prof. Andrea Antal) |
| Wouter Boekel (2013–2017) | "When the brain takes a break: Neural correlates of mind-wandering", University of Amsterdam (co-supervision with Prof. Birte Forstmann) |

*Master/Diploma students*

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| Tarjej Hotvedt (2016) | "Mindfulness and Psoriasis: A pilot study", UiT |
| Are Svend (2014/2015) | "Eksistensiell dynamisk terapi på inneliggende pasienter – en eksplorerende studie av utfall og relasjonelle virkningsmekanismer", UiT (co-supervisor) |
| Ida Marie Opdahl (2014/2015) | "Wandering body, wandering mind? The relationship between bodily movement, creativity and mind-wandering.", University of Tromsø |
| Ira Beckmann (2010) | "Negatives Priming als Identifikations- oder Selektionsphänomen", University of Göttingen |

*Research Assistents*

| | |
|---|---|
| Divyaratan Popli | Eye-tracking and pupillometic correlates of mind wandering (2013, University of Amsterdam) |
| Isabel Viola Kreis | The role of executive functions in mind wandering (2016, University of Tromsø) |
| Carole Schmidt | The role of executive functions in mind wandering (2017, University of Tromsø) |

*Bachelor students*

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|---|---|
| Christian Fossheim (2015) | "Moment-to-moment fluctuations in emotional valence and mind-wandering" |
| Ingrid Marie Skjerstad (2017) | "The wandering mind of the long-distance runner" |
| Anna Elfrida Vollsund (2018) | "Effectiveness of HD-tDCS on influencing executive control during mind wandering" (`https://osf.io/nm2sz/`) |
| Iselin Finn (2018) | "Effectiveness of HD-tDCS on influencing executive control during mind wandering" (`https://osf.io/nm2sz/`) |

## Grants

*HelseNord project PFP1237_15* 2015–2018
3 million NOK (≈ 350.000 EUR)
Optimizing transcranial direct current stimulation for treatment of chronic pain and depression
(3 year full-time post-doc project)

*PhD candidate from Health faculty, UiT* 2016–2020
Neural and computational correlates of mind wandering
(4 year PhD project)

## Personal Scholarships & Awards

*Incentive grant* 2017
Grant awarded by the Faculty of Health for excellent grades in external applications (200k NOK), University of Tromsø

*Incentive grant* 2016
Grant awarded by the Faculty of Health for excellent grades in external applications (200k NOK), University of Tromsø

*Research Prize* 2015
Research prize from the Institute for psychology, University of Tromsø

*GGNB Fellowship* 2009/05–2012/01
Full–Time PhD Scholarship offered by Göttingen Graduate School for Neurosciences and Molecular Biosciences (GGNB)

*GGNB Bridging Fund Scholarship* 2008/10–2008/12
Scholarship offered by Göttingen Graduate School for Neurosciences and Molecular Biosciences (GGNB)

*DFG Scholarship* 2009/01–2009/04
Full–Time Scholarship "Passungsverhältnisse schulischen Lernens: Verstehen und Optimieren"

*Erasmus Scholarship* 2004/07–2005/08
Scholarship for studies abroad, Oslo (UiO), Norway

## Skills

*Experimental Methods*
fMRI, EEG/ERP, EOG, tDCS, eye-tracking, pupil diameter

*Programming Languages*
Very Good: Python, R, C, C++, Matlab/Octave
Good: Perl, Java, SQL, PHP, Presentation
Moderate: F77, Assembler, Objective C, Bash, Javascript

*Computing*
Parallel Programming (MPI), GPU Programming (OpenGL), Unix/Linux Administration, LaTeX, HTML/CSS

**Languages**

*German* (native)
*English* (fluently written, read and spoken)
*Norwegian* (fluently written, read and spoken)
*French* (basic skills)
*Dutch* (basic skills)

**Sports**

running, skiing, judo, rock-climbing, hiking, table-tennis, football